Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle
MATHS TUTORIALS



Half Past Paper – Week 12

Higher Tier

Total Marks

Q	Topic		Sparx Code
2	Compound interest calculations	3	U332
4	Interpreting line graphs	2	U193
6	Finding the surface area of cylinders	5	U464
8a	Using Pythagoras' theorem in 2D	1	U385
8b	Enlargement by a positive or negative scale factor	1	U134
10	Percentage change	3	U286,U278
12	Indices of the form 1/a	1	U985
14	The area rule	4	U592
16	Position-to-term rules for quadratic sequences	3	U206
18a	Expanding triple brackets	3	U606
18b	Solving quadratic inequalities	3	U133
20	Solving simultaneous equations involving quadratics	5	U547
22	Equations of circles and tangents, Finding unknown sides in right-angled triangles	4	U567,U283
	Total	38	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

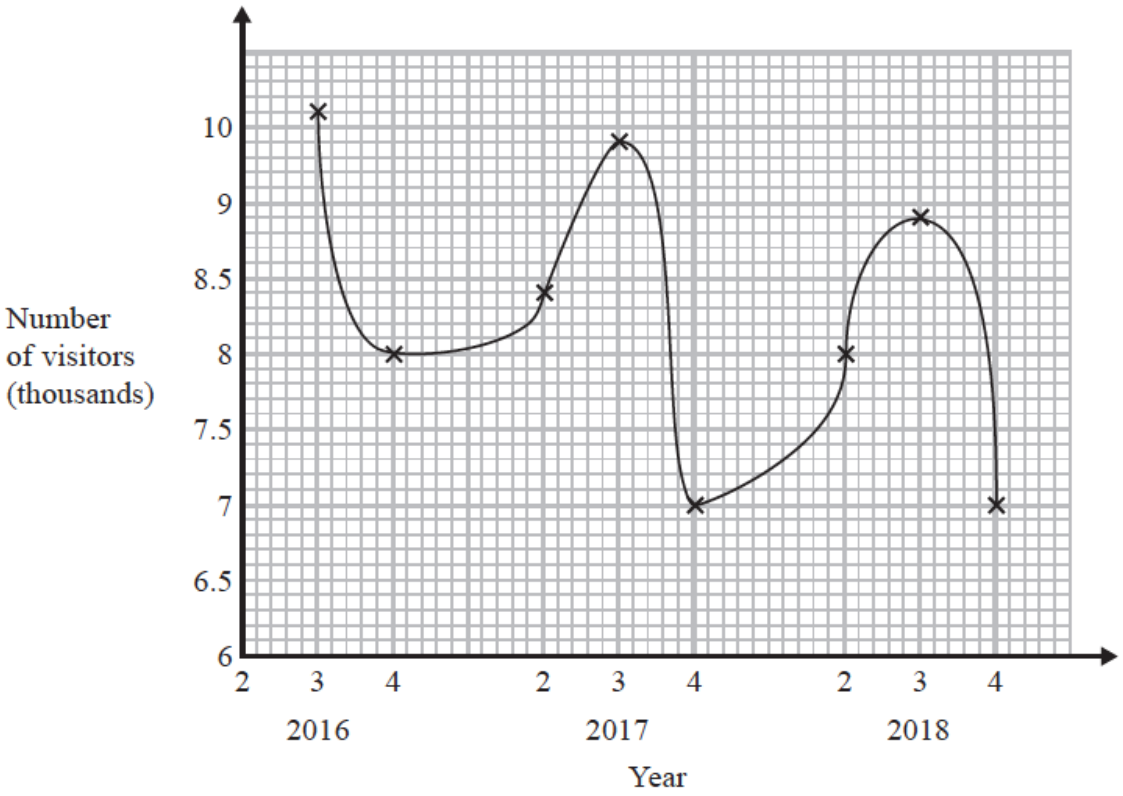
2 Katy invests £200 000 in a savings account for 4 years.
The account pays compound interest at a rate of 1.5 % per annum.

Calculate the total amount of interest Katy will get at the end of 4 years.

£.....

(Total for Question 2 is 3 marks)

4 Sean has drawn a time series graph to show the numbers, in thousands, of visitors to a fun park.



Write down two things that are wrong or could be misleading with this graph.

1 .....

.....

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2 .....

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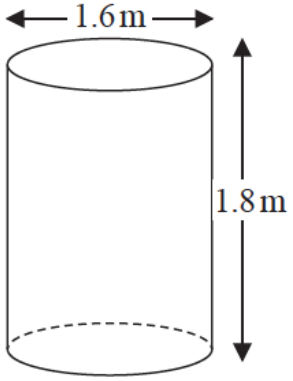
(Total for Question 4 is 2 marks)

6 Jeremy has to cover 3 tanks completely with paint.

Each tank is in the shape of a cylinder with a top and a bottom.
The tank has a diameter of 1.6 m and a height of 1.8 m.

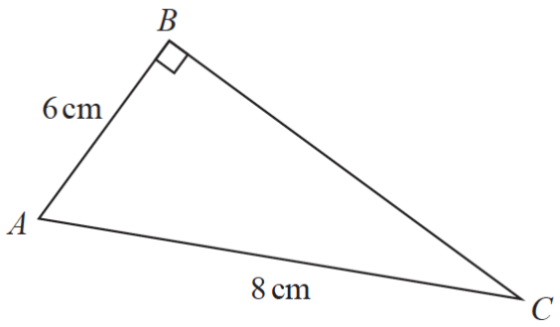
Jeremy has 7 tins of paint.
Each tin of paint covers 5 m^2

Has Jeremy got enough paint to cover completely the 3 tanks?
You must show how you get your answer.



(Total for Question 6 is 5 marks)

8 *ABC* is a right-angled triangle.



Here is Sarah's method to find the length of *BC*.

$$BC^2 = AB^2 + AC^2$$
$$= 6^2 + 8^2$$
$$= 100$$
$$BC = 10$$

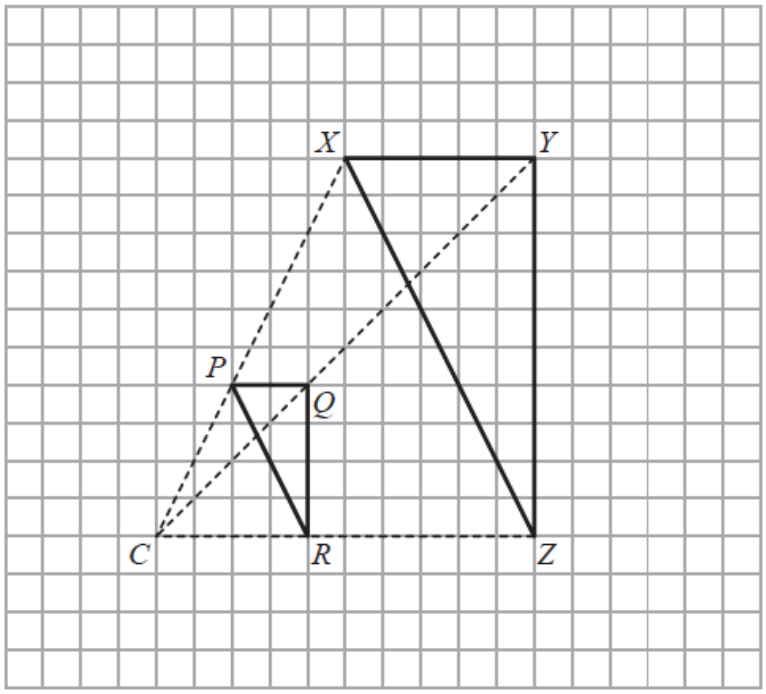
(*a*) What mistake has Sarah made in her method?

.....

.....

.....

(1)



Roy is going to enlarge triangle *PQR* with centre *C* and scale factor $1\frac{1}{2}$

He draws triangle *XYZ*.

(*b*) Explain why Roy's diagram is **not** correct.

.....

.....

.....

(1)

(Total for Question 8 is 2 marks)

10 Marie invests £8000 in an account for one year.
At the end of the year, interest is added to her account.

Marie pays tax on this interest at a rate of 20%
She pays £28.80 tax.

Work out the percentage interest rate for the account.

..... %

(Total for Question 10 is 3 marks)

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12 Patrick has to work out the exact value of $64^{\frac{1}{4}}$

Patrick says,

"$\frac{1}{4}$ of 64 is 16 so $64^{\frac{1}{4}} = 16$ "

Explain what is wrong with what Patrick says.

.....

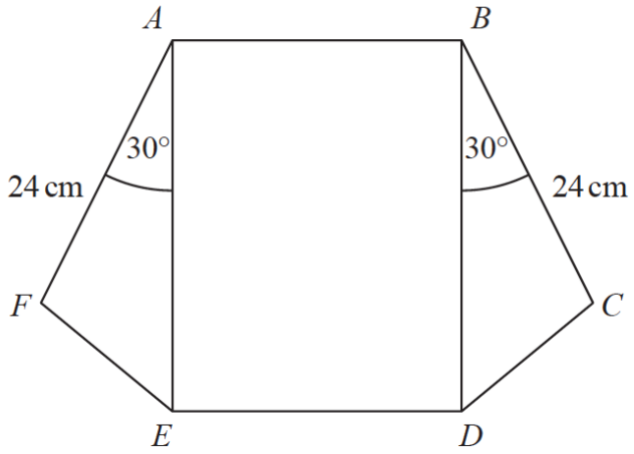
.....

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(Total for Question 12 is 1 mark)

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14 The diagram shows a rectangle, *ABDE*, and two congruent triangles, *AFE* and *BCD*.



area of rectangle *ABDE* = area of triangle *AFE* + area of triangle *BCD*

$$AB : AE = 1 : 3$$

Work out the length of *AE*.

..... cm

(Total for Question 14 is 4 marks)

16 Here are the first six terms of a quadratic sequence.

$$-1 \qquad 5 \qquad 15 \qquad 29 \qquad 47 \qquad 69$$

Find an expression, in terms of n, for the nth term of this sequence.

.....

(Total for Question 16 is 3 marks)

18 (*a*) Show that $(2x + 1)(x + 3)(3x + 7)$ can be written in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are integers.

(3)

(*b*) Solve $(1 - x)^2 < \frac{9}{25}$

.....

(3)

(Total for Question 18 is 6 marks)

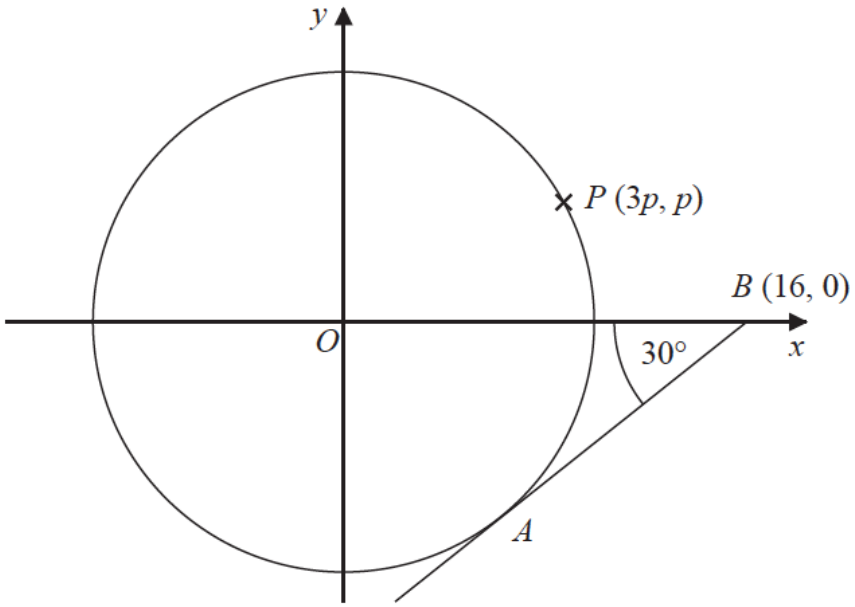
20 Solve algebraically the simultaneous equations

$$x^2 - 4y^2 = 9$$
$$3x + 4y = 7$$

.....

(Total for Question 20 is 5 marks)

22 The diagram shows a circle, centre O.



AB is the tangent to the circle at the point A.
Angle $OBA = 30°$

Point B has coordinates (16, 0)
Point P has coordinates $(3p, p)$

Find the value of p.
Give your answer correct to 1 decimal place.
You must show all your working.

$p =$ .....

(Total for Question 22 is 4 marks)

TOTAL FOR PAPER IS 41 MARKS